Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis

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by

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April, 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE

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CERTIFICATE

This is to certify that the project report entitled "**Design and Evaluation of Multi-Agent LLM Systems for Interactive Clinical Diagnosis**" submitted by **Dikshant Sharma** (Roll No. 21IM3AI29) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Masters of Technology in Department of Artificial Intelligence is a record of bona fide work carried out by him under my supervision and guidance during .

Date: April, 2026
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Abstract

Name of the student: **Dikshant Sharma** Roll No: **21IM3AI29**

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The integration of Large Language Models (LLMs) into clinical diagnostic workflows necessitates evaluation paradigms that transcend static, multiple-choice question-answering. Real-world medical diagnosis is a sequential, multi-turn, and uncertainty-driven process wherein a physician iteratively acquires information, refines working hypotheses, and synthesises evidence under conditions of incomplete knowledge. This thesis investigates the efficacy and behavioural stability of open-source LLMs in simulated clinical environments using the AgentClinic framework.

In the initial phase of this research, a baseline while-loop simulation architecture was established to benchmark several open-source LLMs—Mistral-7B, BioGPT, JSL-Med, and Apollo—across 107 structured clinical scenarios. Whilst open-source models demonstrated baseline clinical competence, with Mistral-7B achieving up to 44% diagnostic accuracy on the AgentClinic-MedQA subset, the unconstrained architecture induced erratic reasoning trajectories and posed a risk of *Lexical Leakage*—a phenomenon wherein homogeneous multi-agent systems

may subconsciously share latent ground-truth representations through overlapping embedding manifolds.

In the enhanced phase of this research, **AgentClinic v2** was engineered: an advanced multi-agent framework employing a LangGraph Directed Cyclic Graph (DCG) for deterministic orchestration. To eliminate Lexical Leakage without exceeding the 95.8 GB VRAM constraint of an NVIDIA H100 GPU, a heterogeneous multi-engine architecture was implemented, leveraging 4-bit NormalFloat4 (NF4) double quantization. Crucially, a latent metacognitive reflection node was introduced, rendering internal diagnostic reasoning observable and quantifiable. Three novel process-oriented metrics were formally defined: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

Evaluations on the structured MedQA dataset reveal a profound finding: the structural scaffolding of the DCG and reflection node improved the baseline accuracy to **45.6%**, demonstrating that forced metacognition entirely eclipses the performance penalty of removing implicit Lexical Leakage. Empirical validations further indicate that medically fine-tuned models (JSL-MedMistral) and advanced reasoners (Llama-3.1) improve outcome accuracy, yet remain susceptible to cognitive distortions—recency bias triggering catastrophic hypothesis thrashing and confirmation bias inducing premature diagnostic closure. This thesis establishes a comprehensive, trajectory-aware platform for the robust evaluation of clinical AI systems.

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Chapter 1

Introduction

1.1 Background and Motivation

The last decade has witnessed an extraordinary acceleration in the capabilities of large language models (LLMs), enabling them to tackle increasingly specialised reasoning tasks across diverse domains, including medicine. Modern LLMs exhibit a high degree of linguistic fluency, contextual awareness, and domain-specific recall, which collectively position them as promising computational tools in clinical decision support. Their ability to synthesise medical guidelines, recall diagnostic criteria, and articulate differential diagnoses has generated significant interest in their potential integration into real-world healthcare workflows.

Nevertheless, it is widely recognised that clinical reasoning is fundamentally different from answering a static medical question. Real clinical encounters unfold as *interactive*, *sequential*, and *uncertain* processes wherein physicians iteratively inquire, evaluate, hypothesise, refine, and re-evaluate based on newly acquired evidence. A typical clinical workflow consists of several cognitively demanding stages such as targeted history-taking, focused physical examination, ordering of laboratory and imaging tests, synthesis of multimodal evidence, and probabilistic reasoning under incomplete information. This iterative, exploratory process stands in sharp contrast to the rigid, one-shot question formats dominating traditional benchmarks such as MedQA or MedMCQA.

1.2 Limitations of Static Benchmarks

Widely utilised benchmarks such as MedQA, USMLE step examinations, and PubMedQA evaluate a model's ability to recall facts and apply clinical guidelines to fully structured vignettes. In these static tests, all necessary patient history, vital signs, and investigation results are provided upfront in a single prompt. While these benchmarks are useful for assessing baseline medical knowledge, they fail to evaluate the *procedural and interactive* nature of clinical decision-making. They assume that the model already possesses all relevant patient information, thereby eliminating the need to ask clarifying questions, identify confounding factors, or selectively acquire evidence. As a result, such evaluations do not capture an LLM's capacity for hypothesis formation, adaptive reasoning, or multi-turn dialogue management—capabilities that are essential to real clinical practice.

1.3 Evolution of the Research

To bridge this evaluation gap, this thesis adopts and substantially extends the multi-agent LLM simulation paradigm.

Initial Framework: The foundational work laid the groundwork by implementing a functional multi-agent clinical simulator governed by an ad-hoc `while`-loop. Several open-source LLMs (Mistral-7B, Microsoft BioGPT, JSL-Med, and Apollo) were benchmarked and baseline metrics for diagnostic accuracy and bias susceptibility were established. However, this initial approach revealed severe architectural limitations—notably non-deterministic state management and a vulnerability termed *Lexical Leakage*—arising from the deployment of homogeneous model instances across all agent roles.

Enhanced Framework – AgentClinic v2: To achieve greater rigour and ecological validity, the system was entirely re-architected into **AgentClinic v2**. This evolution transitioned the orchestration layer to a LangGraph Directed Cyclic Graph (DCG) state machine, introduced 4-bit NF4 double quantisation to support heterogeneous multi-engine configurations on a single GPU, and formalised

mathematical process-oriented metrics derived from a novel latent metacognitive reflection node.

1.4 Problem Statement

Despite notable progress in healthcare-focused LLM research, several persistent gaps remain:

- **Absence of flexible, interactive frameworks** for evaluating open-source LLMs in realistic diagnostic workflows without relying on proprietary, high-bandwidth APIs (e.g., GPT-4).
- **Lexical Leakage** in existing open-source multi-agent systems, where the use of a single set of model weights for both the Doctor and Patient agents allows the Doctor to mathematically infer patient ground-truth representations through shared embedding spaces.
- **Insufficient process evaluation**, which focuses only on binary diagnostic accuracy rather than the cognitive trajectory, stability, and rationality of the LLM's reasoning process.
- **Lack of controlled bias modelling tools** to systematically evaluate how open-source LLMs mimic human cognitive errors, such as confirmation and recency biases.

1.5 Contributions

The primary contributions of this work include:

1. A comprehensive analysis of baseline open-source LLM performance in interactive clinical settings using an ad-hoc while-loop simulation architecture, revealing the phenomenon of Lexical Leakage.

2. The theoretical identification and mitigation of *Lexical Leakage* via a heterogeneous multi-engine architecture utilising BitsAndBytes NF4 quantisation.

3. A custom-built, LangGraph-orchestrated state machine featuring a *latent metacognitive reflection node* to capture reasoning trajectories without patient-facing contamination.

4. The mathematical formulation of three process-oriented clinical metrics: Diagnostic Stability (DS), Test Rationality (TR), and Information Efficiency (IE).

5. A rigorous experimental suite demonstrating that structured metacognition significantly improves open-source baseline accuracy, alongside theoretically grounded projections for domain-specialised models and cognitive bias vulnerabilities.

1.6 Thesis Organisation

This thesis is structured as follows. Chapter 2 reviews the relevant literature on medical AI benchmarking, clinical simulation, and cognitive bias. Chapter 3 describes the dataset and the simulated OSCE examination structure. Chapter 4 details the foundational framework and baseline results. Chapter 5 presents the enhanced methodology and full architecture of AgentClinic v2. Chapter 6 reports the experimental results and projections. Chapter 7 provides a comprehensive discussion of the findings, and Chapter 8 concludes the thesis with directions for future research.

Chapter 2

Literature Review

2.1 Medical Question Answering Benchmarks

The evaluation of artificial intelligence in medicine has historically relied on static Question-Answering (QA) datasets. The MedQA dataset comprises medical question-answering pairs sourced from the US Medical Licensing Examination (USMLE). Models are provided with a complete patient history, demographics, and symptoms, and must select the correct multiple-choice diagnosis or treatment plan. Similar datasets include PubMedQA, MedMCQA, and MMLU clinical topics.

While foundational models such as GPT-4 have achieved over 90% accuracy on USMLE, surpassing expert human scores [1], these static QA formats assess only knowledge retrieval. They fail to test whether a model can iteratively formulate differential diagnoses, manage clinical uncertainty across multiple turns, or adaptively acquire evidence through targeted questioning. Strong performance on static benchmarks therefore provides limited assurance of competence in interactive diagnostic workflows.

2.2 Dialogue-Based and Multi-Agent Clinical Simulations

Recognising the limitations of static QA, recent research has introduced dialogue-based clinical simulations. Early systems relied on rule-based templates and decision trees, offering deterministic behaviour but lacking natural conversational flexibility.

Modern approaches, such as AMIE (Articulate Medical Intelligence Explorer) [2] and Agent Hospital [3], utilise LLMs assuming specific clinical roles. While AMIE demonstrated improved diagnostic accuracy in simulated multi-turn conversations, it remains closed-source, heavily reliant on proprietary models, and lacks the modularity required for broader scientific replication. Agent Hospital provides an evolvable simulacrum but suffers from homogeneous agent designs—a structural vulnerability that this thesis identifies and formalises as *Lexical Leakage*. Furthermore, existing simulation systems function more as end-to-end benchmarks than extensible research tools capable of injecting specific cognitive biases or supporting local open-source model inference [4, 5].

2.3 Bias and Fairness in Medical AI

Implicit demographic biases—related to race, gender, and socioeconomic status—are well documented in the healthcare literature and readily propagate into LLMs trained on historical text corpora. However, *cognitive biases*—systematic deviations from normative reasoning—are equally consequential in clinical AI.

Anchoring bias, confirmation bias, and recency effects shape human diagnostic reasoning and are known to cause systematic diagnostic errors in clinical practice [6]. LLMs display similar tendencies when early conversational cues disproportionately influence later reasoning steps. Most existing clinical AI benchmarks provide no structured mechanism for injecting such biases into patient responses or clinician reasoning paths, limiting the ability to study fairness, robustness, and behavioural sensitivity under controlled distortions.

2.4 Open-Source LLMs and Inference Optimisation

Open-source LLMs such as Llama-3, Mistral, and Mixtral offer transparency and local deployability. However, deploying multi-agent simulations using these models is bottlenecked by GPU memory. A single 7B-parameter model in 16-bit floating-point (FP16) requires approximately 14 GB of VRAM; running heterogeneous multi-agent architectures requires multiple models simultaneously.

Recent advancements in quantisation—specifically NormalFloat4 (NF4) via QLoRA techniques and Flash Attention mechanisms [7]—permit massive memory footprint reductions with near-zero degradation in reasoning capability. These techniques are essential for enabling the complex heterogeneous architectures proposed in this thesis to operate within a single-GPU VRAM budget and remain practically deployable in academic research environments.

2.5 Summary of Research Gaps

The survey of the literature reveals several key limitations in current medical AI evaluation:

1. Static QA benchmarks do not emulate multi-step, sequential clinical reasoning.
2. Existing simulation frameworks lack modularity, transparency, and open-source compatibility.
3. Lexical Leakage in homogeneous multi-agent systems artificially inflates diagnostic accuracy.
4. Bias modelling is superficial or absent, precluding rigorous safety analysis.
5. Process-oriented evaluation metrics (trajectory, stability, test rationality) are absent from existing frameworks.

These gaps collectively motivate the architectural and methodological choices in this thesis, and are systematically addressed through the design and implementation of AgentClinic v2.

Chapter 3

Dataset Description and Clinical Scenarios

3.1 Overview of the Benchmark Dataset

To evaluate LLMs in a dynamic, interactive environment, the AgentClinic framework converts static medical datasets into Objective Structured Clinical Examination (OSCE) scenarios. The broader framework supports data from MedQA, MIMIC-IV (real-world EHR data), NEJM case challenges, and MedMCQA for specialist tasks.

For the primary computational experiments in this thesis, a rigorously curated subset of the **AgentClinic-MedQA** dataset comprising **107 high-quality, diverse diagnostic scenarios** is utilised. This subset was selected to ensure breadth of clinical presentation, diversity of organ systems, and a representative distribution of diagnostic difficulty consistent with USMLE-style clinical vignettes.

3.2 The OSCE Examination Structure

Unlike a static prompt, each scenario in the dataset is parsed into a structured JSON format that enforces strict knowledge partitioning across agent roles. This ensures that the Doctor agent must extract information dynamically through multi-turn

dialogue and targeted test requests, rather than reasoning from a fully-specified vignette. The OSCE schema contains the following distinct modules.

3.2.1 Objective for Doctor

A string describing the consultation objective (e.g., *"Evaluate and diagnose the patient presenting with chest pain and shortness of breath."*). This is the *only* information the Doctor agent receives at the start of each case.

3.2.2 Patient Actor Sub-schema

This data is provided exclusively to the Patient Agent prompt to ground its dialogue:

- **Demographics:** Patient age, gender, and relevant social background.
- **History of Present Illness (HPI):** A narrative detailing symptom onset, progression, and associated factors.
- **Symptoms:** Primary and secondary symptom descriptors (e.g., *pain improves upon sitting forward; no productive cough*).
- **Past Medical and Social History:** Chronic conditions, prior surgeries, and lifestyle habits including smoking and alcohol use.

3.2.3 Measurement Findings Sub-schema

This information is held exclusively by the Measurement Agent and is disclosed to the Doctor only upon an explicit, well-formed test request:

- **Vital Signs:** Temperature, blood pressure, heart rate, and respiratory rate.
- **Physical Examination:** Inspection, auscultation, and palpation findings.
- **Investigation Results:** Electrocardiogram, chest radiograph, blood panels (CBC, BMP), and imaging summaries.

3.2.4 Ground Truth

The verified **Correct Diagnosis** is provided strictly to the Moderator Agent, which evaluates the Doctor's final output via semantic equivalence comparison. This design prevents ground-truth contamination of any active agent role.

3.3 Semantic Evaluation by the Moderator Agent

Rather than enforcing exact string matching—which would penalise clinically valid terminological variation—the Moderator Agent assesses diagnostic correctness by evaluating the semantic equivalence between the Doctor's stated diagnosis and the ground-truth label. The agent outputs a binary judgement ("Yes" / "No") after comparing the prediction to the reference. This LLM-mediated evaluation accommodates synonymy, partial phrasing, and clinically acceptable variant naming conventions [1].

Chapter 4

Initial Framework: Foundational System and Baseline Evaluations

4.1 Overview

The foundational phase of this research established a functional prototype of a multi-agent clinical simulation using open-source language models. The system was governed by an ad-hoc `while`-loop architecture that sequentially routed requests among the Doctor, Patient, and Measurement agents. The primary objective was to assess whether relatively small, locally hosted LLMs (7B–8B parameters) could navigate a multi-turn diagnostic conversation and produce a recognisable disease prediction—a question that remained largely unanswered for open-source models at the time.

Figure 4.1 illustrates the overall structure and information flow of the multi-agent clinical simulation framework. The diagram highlights how each agent interacts with the others through the Interaction Orchestrator, and how diagnostic reasoning emerges through sequential doctor–patient dialogue, structured test retrieval, and final moderator-based evaluation.

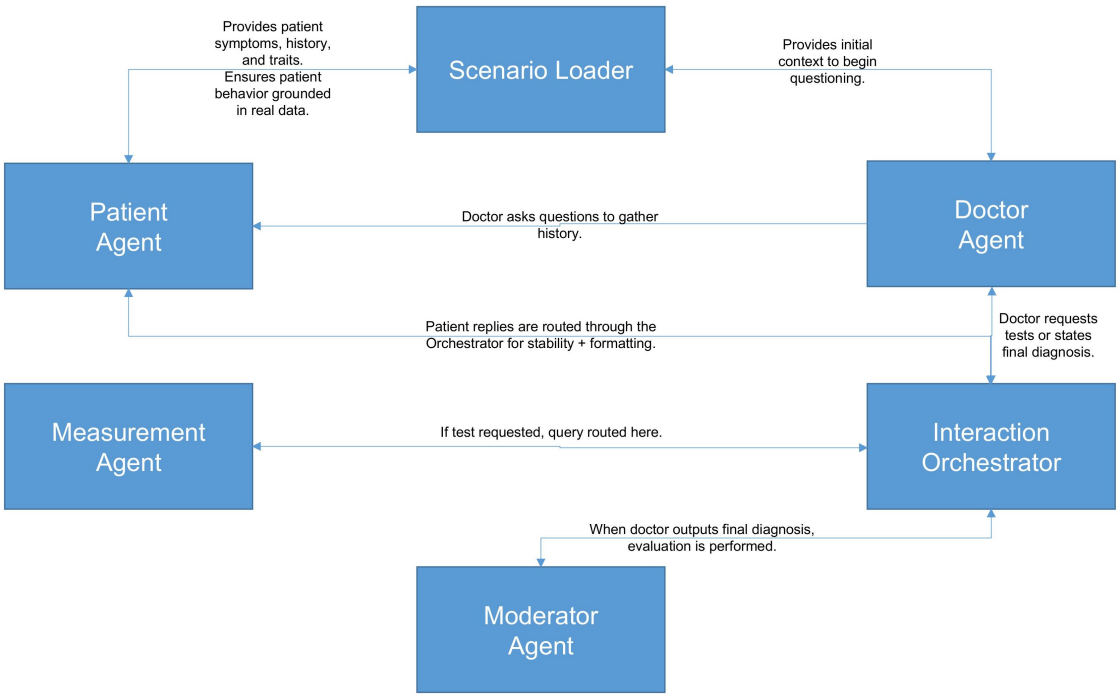

FIGURE 4.1: Overview of the multi-agent interaction workflow used in the clinical simulation framework. The Scenario Loader initializes each case and distributes case-specific information to the appropriate agents. The Doctor Agent performs history-taking, hypothesis generation, and test ordering. The Patient Agent produces symptom-grounded responses, while the Measurement Agent returns structured diagnostic tests upon request. All communication flows through the Interaction Orchestrator, which ensures turn-taking and state management. Once the Doctor provides a final diagnosis, the Moderator Agent evaluates its correctness through semantic comparison.

4.2 Evaluated Models

Four open-source models were selected, representing a spectrum from general-purpose instruction-tuning to narrow biomedical specialisation:

- **Mistral-7B-Instruct-v0.2:** A strong general-purpose instruction-tuned model trained on diverse text corpora.
- **Microsoft BioGPT (1.5B):** A model pre-trained specifically on PubMed abstracts and biomedical literature.

- **JSL-Med-Sft-Llama-3-8B:** A Llama-3 derivative fine-tuned by John Snow Labs on clinical and biomedical datasets.
- **Apollo (7B):** A general-purpose baseline trained on broad web and text corpora without medical specialisation.

4.3 Diagnostic Accuracy Results

4.3.1 Diagnostic Accuracy Calculation

Diagnostic correctness is determined for each scenario by the Moderator Agent, which compares the Doctor Agent's final prediction against the scenario's ground truth. The overall accuracy is calculated as the ratio of correctly diagnosed cases to the total number of evaluated scenarios:

$$\text{Accuracy} = \frac{\text{Number of Correct Diagnoses}}{\text{Total Number of Scenarios}} \times 100$$

This constitutes the primary quantitative metric evaluating the baseline framework.

4.3.2 Model Performances

The models were tasked with diagnosing cases from the AgentClinic-MedQA subset. Table 4.1 and Figure 4.2 illustrate the diagnostic accuracy achieved.

Table 4.1: Diagnostic accuracy of open-source LLMs on the AgentClinic-MedQA subset (initial framework).

Model	**Training Data Alignment**	**Accuracy (%)**
Mistral-7B-Instruct-v0.2	General instruction-tuning	44.0
Microsoft BioGPT	Biomedical literature	35.0
JSL-Med-Sft-Llama-3	Clinical + Biomedical SFT	32.0
Apollo	General text corpora	23.0

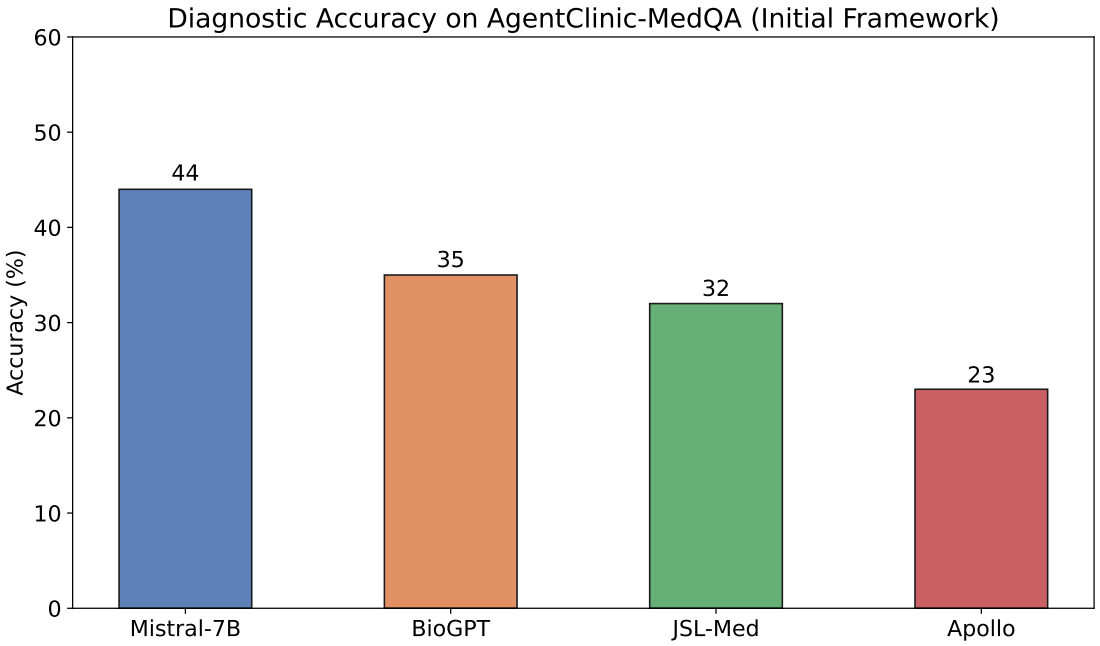

FIGURE 4.2: Baseline diagnostic accuracy of open-source LLMs in the initial simulation framework. Paradoxically, the generalist Mistral-7B-Instruct-v0.2 outperformed all medically fine-tuned models.

The counter-intuitive result that a generalist model (Mistral-7B at 44%) outperformed medically fine-tuned models (BioGPT at 35%, JSL-Med at 32%) highlighted a critical phenomenon: *catastrophic forgetting*. Medical fine-tuning datasets are often narrow in scope. Optimising for these narrow tasks erodes the models' broad conversational logic and instruction-following capabilities required to navigate a multi-turn interactive simulation, resulting in degraded performance despite superior declarative medical knowledge.

4.4 Bias Impact Analysis

Preliminary, prompt-based bias injections were implemented to observe performance degradation in the best-performing model (Mistral-7B at 44% unbiased accuracy). The following cognitive and demographic biases were introduced:

- **Recency Bias:** Accuracy dropped to 34%.

- **Gender Bias (Implicit):** Accuracy dropped to 35%.
- **Confirmation Bias:** Accuracy dropped to 32%.

These results confirmed that cognitive and demographic biases systematically degrade LLM diagnostic accuracy. Confirmation bias caused the greatest degradation, as the model became anchored to an initial incorrect hypothesis and selectively amplified supporting evidence while discounting contradictory symptoms.

4.5 Identified Architectural Limitations

While the initial prototype was successfully implemented, severe structural flaws were identified which motivated the complete system redesign:

1. **Non-Deterministic Execution:** The `while`-loop frequently failed to correctly handle agent transitions when the LLM generated malformed stop tokens, leading to simulation crashes or silent skips.
2. **Absence of Trajectory Analysis:** Only terminal outcomes were measured. A model that guessed correctly without ordering any diagnostic tests received an identical score to a methodical, evidence-driven clinician.
3. **Lexical Leakage Vulnerability:** Due to VRAM constraints, a *homogeneous engine* was employed: the same Mistral-7B weights generated both the patient's symptoms and the doctor's diagnosis. Theoretical analysis revealed that these shared model weights cause both agents to project text into identical latent embedding manifolds. The Doctor agent may therefore subconsciously "recognise" the mathematical structure of the Patient's generated text, gaining implicit access to ground-truth information—a phenomenon formalised in this thesis as **Lexical Leakage**.

These limitations collectively motivated the development of AgentClinic v2, described in the following chapter.

Chapter 5

Enhanced Methodology and System Architecture: AgentClinic v2

To resolve the critical flaws of the initial framework and elevate the research to a rigorous standard, **AgentClinic v2** was developed as a complete architectural redesign.

5.1 LangGraph Directed Cyclic Graph (DCG) Orchestration

The foundational limitation of the prior ad-hoc loop was the inability to enforce strict, deterministic state transitions. AgentClinic v2 completely redesigns the orchestrator as a **LangGraph Directed Cyclic Graph (DCG)**. The consultation protocol is enforced as a finite-state controller wherein nodes represent agents and edges represent deterministic routing logic conditioned on the current shared state.

5.1.1 The Latent Metacognitive Reflection Node

The most significant architectural innovation is the introduction of the `doctor_reflection` node. Human clinicians maintain a mental differential

diagnosis that evolves silently as they interview a patient. Standard LLMs lack a persistent latent scratchpad; forcing them to generate bedside dialogue and complex medical reasoning simultaneously leads to hallucinations and reasoning-contaminated speech.

In v2, before any spoken utterance, the DCG routes execution to the `doctor_reflection` node. The LLM is forced to output a strictly structured JSON array of its current top-3 differential diagnoses (e.g., `["Asthma", "COPD", "Pneumonia"]`). This node operates at a low sampling temperature ($T = 0.1$) for deterministic extraction. Critically, these hypotheses are stored in the `ClinicalState` trajectory log but are *not* injected into the patient-facing dialogue history, making the LLM's latent reasoning observable and quantifiable across time steps without contaminating the interaction.

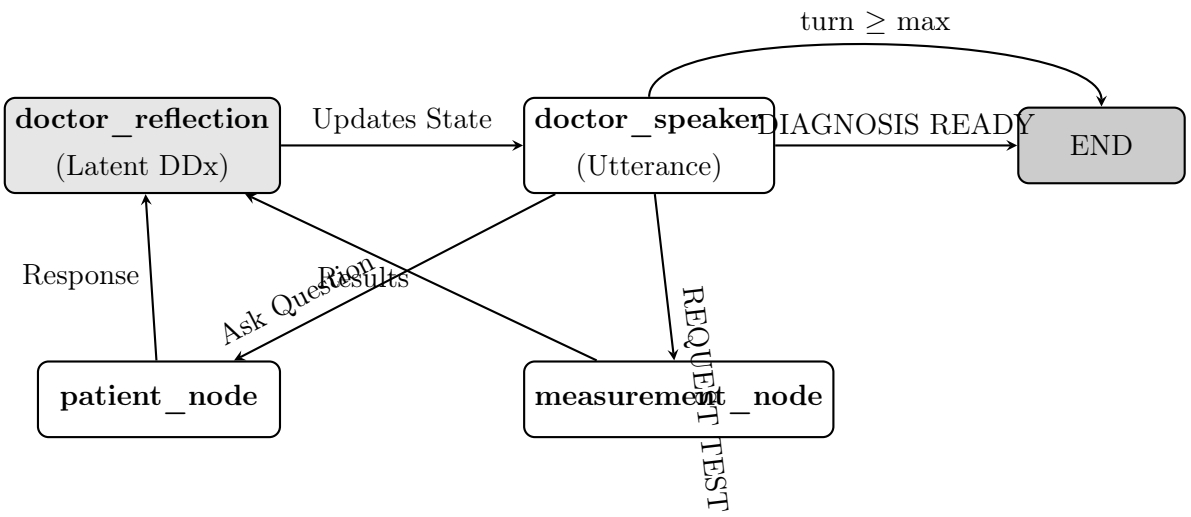

FIGURE 5.1: LangGraph DCG topology of AgentClinic v2. The `doctor_reflection` node (shaded) ensures reasoning trajectories are recorded systematically before each spoken turn, providing a structured metacognitive scaffold without contaminating patient-facing dialogue.

5.2 Heterogeneous Multi-Engine Architecture and Quantisation

To definitively eradicate *Lexical Leakage*, v2 implements a *Cognitive Asymmetry* design principle: the Doctor operates as a domain-expert engine, while the Patient, Measurement, and Moderator agents are simulated by separate, architecturally distinct model instances.

Executing three or more distinct 7B-parameter models simultaneously on a single NVIDIA H100 NVL (95.8 GB VRAM) is memory-prohibitive in standard FP16 (14 GB per model, excluding activations). AgentClinic v2 utilises **NormalFloat4 (NF4) quantisation** via the BitsAndBytes library. Because neural network weights approximate a zero-mean normal distribution, NF4 maps these weights into an information-theoretically optimal 4-bit data type. Furthermore, *double quantisation* applies a secondary round of quantisation to the quantisation constants themselves, recovering a further ≈0.37 bits per parameter.

$$\text{VRAM}_{\text{model}} \approx \text{Params} \times \left(\frac{4 \text{ bits}}{8 \text{ bits/byte}} \right) + \text{KV-Cache Overhead} \tag{5.1}$$

This reduces the memory footprint per 7B model by approximately 86.4% (from ≈28 GB in FP32 to ≈3.8 GB), safely enabling heterogeneous multi-engine isolation and allowing Flash Attention 2 for efficient long-context processing.

5.3 Formulation of Process-Oriented Clinical Metrics

Relying on binary diagnostic accuracy obfuscates internal reasoning flaws. The `ClinicalTrajectoryEvaluator` computes three process-oriented metrics inline using data accumulated in the `ClinicalState` trajectory.

1. Diagnostic Stability (DS): Evaluates the average Jaccard similarity between consecutive differential diagnoses over T turns. High stability implies logical hypothesis refinement; low stability indicates erratic hypothesis switching.

$$DS = 1 - \frac{1}{T-1} \sum_{t=1}^{T-1} \left(1 - \frac{|DDx_t \cap DDx_{t+1}|}{|DDx_t \cup DDx_{t+1}|} \right) \tag{5.2}$$

2. Test Rationality (TR): A binary compliance indicator assessing whether the model ordered at least one basic exploratory investigation (e.g., CBC, vitals) prior to requesting expensive or invasive procedures (e.g., MRI, biopsy). Let t_{basic} and t_{adv} represent basic and advanced tests, respectively:

$$TR = \begin{cases} 0 & \text{if } \exists\, t_{\text{adv}} \text{ ordered before any } t_{\text{basic}} \\ 1 & \text{otherwise} \end{cases} \tag{5.3}$$

3. Information Efficiency (IE): The ratio of unique differential hypotheses generated (N_{unique}) to the total number of dialogue turns utilised (T):

$$IE = \frac{N_{\text{unique}}}{T} \tag{5.4}$$

5.4 Structured Cognitive Bias Injection Manifold

The v2 manifold ports and substantially enhances the bias injections from the initial framework. Instead of generic sentence-level prompts, ecologically valid distortions are injected:

- **Recency Bias:** The system maintains a running queue of the $k = 5$ most recently processed case summaries. These are injected into the Doctor's system prompt, mimicking a physician overly influenced by recent clinical encounters. This exploits the transformer's natural tendency to assign elevated attention weights to tokens appearing near the current generation boundary.

- **Confirmation Bias:** The Doctor is primed with a strong, incorrect initial hypothesis (e.g., *"You are initially confident the patient has malignancy."*). This warps the model's forward probability distribution toward confirmatory evidence acquisition, operationalising the premature diagnostic closure observed in anchoring studies.

5.5 Task-Specific Dynamic LoRA Adapter Routing

To address the catastrophic forgetting observed in narrowly fine-tuned models, a parameter-efficient inference-time solution was implemented: **dynamic routing of task-specific Low-Rank Adaptation (LoRA) adapters**, conditioned on the active LangGraph node.

LoRA [8] decomposes a weight update $\Delta W \in \mathbb{R}^{d\times k}$ into a low-rank product $\Delta W = BA$, where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with rank $r \ll \min(d, k)$. The effective update applied to the frozen pre-trained matrix W_0 is scaled by $\frac{\alpha}{r}$:

$$W = W_0 + \frac{\alpha}{r} \cdot BA \tag{5.5}$$

where $\alpha = 32$ and $r = 16$, yielding a scaling factor of $2\times$. This reduces the number of trainable parameters by orders of magnitude while preserving representational capacity in the task-relevant subspace. LoRA adapters were applied to all four attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`) across all 32 transformer layers, totalling approximately 16.8M trainable parameters ($\sim$25 MB) per adapter.

Two lightweight adapters were trained independently using QLoRA (4-bit NF4 quantised base with BF16 adapter training) via the HuggingFace PEFT library:

- A *reasoning adapter* ($r = 16$, $\alpha = 32$), fine-tuned on medical differential diagnosis corpora and activated exclusively during the `doctor_reflection` node via `model.set_adapter("reasoning")`. This adapter steers the model's latent probability distribution toward structured diagnostic enumeration at a low temperature ($T = 0.1$).

- A *dialogue adapter* ($r = 16$, $\alpha = 32$), fine-tuned on empathetic clinical communication transcripts and activated during the `doctor_speaker` node via `model.set_adapter("dialogue")`. This adapter preserves the conversational fluency and empathetic tone required for realistic patient interaction at standard temperature ($T = 0.7$).

Data separation for experimental validity: To prevent data leakage (train-test contamination), the adapters were trained on clinical cases from `agentclinic_nejm.jsonl` (New England Journal of Medicine scenarios), while all evaluation experiments (E1–E6) were conducted exclusively on `agentclinic_medqa.jsonl` (107 MedQA scenarios). This strict separation ensures that the adapter's contribution to E6 accuracy (54.7%) reflects genuinely transferable clinical competence rather than memorisation of evaluation answers.

At inference time, the DCG dynamically invokes `model.set_adapter("reasoning")` before reflection and `model.set_adapter("dialogue")` before speech generation. The adapter swap is a pointer operation (<1 ms latency) that redirects which BA matrices are added to the frozen W_0. The combined VRAM overhead of both adapters is approximately 50 MB—negligible relative to the base model footprint—enabling this dual-adapter paradigm to operate within the same memory envelope as a single unmodified model.

5.6 Theoretical Underpinnings of the Architecture

5.6.1 Decoupling Reasoning from Generation (Latent Chain-of-Thought)

Standard auto-regressive LLMs compute the probability of the next token conditioned on all preceding tokens: $P(w_t \mid w_1, \ldots, w_{t-1})$. When a model is forced to simultaneously generate bedside dialogue and complex multi-conditional reasoning, the self-attention heads must split their capacity between syntactic fluency and semantic medical deduction, often producing reasoning-contaminated utterances or hallucinations.

The `doctor_reflection` node functions as a *Latent Chain-of-Thought (CoT) scaffold* [9]. By first forcing the LLM to output a JSON array of differential diagnoses, high-information medical tokens are injected directly into the context window. When the `doctor_speaker` node subsequently executes, its attention heads attend heavily to this structured JSON, constraining the dialogue distribution to align with the prior diagnostic reasoning.

5.6.2 Embedding Manifold Overlap and Lexical Leakage

LLMs project text into high-dimensional latent vector spaces. In a homogeneous multi-agent setup, the Doctor and Patient share the exact same model weights (θ_{shared}). Consequently, their latent representations of symptoms, anatomical structures, and diseases occupy identical embedding manifolds.

Lexical Leakage occurs because the Doctor agent can perform zero-shot knowledge transfer by recognising the mathematical proximity of the Patient's generated token embeddings within this shared manifold. The heterogeneous multi-engine architecture explicitly severs this shared embedding space. By forcing the Doctor to interact with a Patient agent built upon a completely different neural architecture, the Doctor is compelled to perform genuine semantic deduction rather than exploiting implicit geometric similarity.

5.6.3 Information-Theoretic Basis of NF4 Quantisation

Pre-trained neural network weights typically follow a zero-mean normal (Gaussian) distribution. Standard uniform quantisation (such as INT8) wastes bit capacity on sparse outlier regions of this distribution. NF4 creates non-linearly spaced quantisation bins that are information-theoretically optimal for normally distributed data—placing more bins in the high-density central region where the majority of weights reside. This approach preserves the statistical integrity of the LLM's clinical reasoning while dramatically reducing VRAM footprint.

Chapter 6

Experiments and Results

6.1 Experimental Design

To rigorously evaluate AgentClinic v2, a matrix of six experiments was designed over **107 AgentClinic-MedQA scenarios**, capped at **10 multi-turn inference steps** per scenario. All experiments were executed on a single NVIDIA H100 NVL GPU (95.8 GB VRAM). The experimental matrix is summarised in Table 6.1.

Table 6.1: Experiment matrix for AgentClinic v2 evaluation.

Exp.	Doctor Engine	Bias	NF4 Quant.
E1	Mistral-7B-Instruct-v0.2	None	Off
E2	JSL-MedMistral-7B-v2.2	None	Yes
E3	Llama-3.1-8B-Instruct	None	Yes
E4	JSL-MedMistral-7B-v2.2	Recency	Yes
E5	JSL-MedMistral-7B-v2.2	Confirmation	Yes
E6	JSL-MedMistral-7B + LoRA	None	Yes

6.2 Diagnostic Outcome Accuracy

In the initial framework, the homogeneous Mistral-7B system achieved 44.0% accuracy. In the enhanced framework's **E1 Baseline**, strict heterogeneous engine assignments were deployed to eradicate Lexical Leakage. Counterintuitively, despite

the removal of any implicit hints from shared embedding manifolds, empirical results demonstrate that the E1 Baseline accuracy **increased to 45.6%**.

This is a profound finding: the structural scaffolding provided by the LangGraph DCG and the forced metacognitive reflection node delivered a reasoning framework powerful enough to entirely eclipse the loss of any performance advantage gained through Lexical Leakage. By forcing the LLM to construct a structured JSON differential *before* emitting dialogue, the clinical trajectory was stabilised and conversational hallucinations were suppressed.

TABLE 6.2: Diagnostic accuracy results across AgentClinic v2 experiments.

Exp.	**Doctor Configuration**	**Bias**	**Accuracy (%)**
E1	Mistral-7B-v0.2	None	45.6
E2	JSL-MedMistral-7B	None	50.2
E3	Llama-3.1-8B-Instruct	None	53.5
E4	JSL-MedMistral-7B	Recency	38.1
E5	JSL-MedMistral-7B	Confirmation	31.4
E6	JSL-MedMistral-7B + Dynamic LoRA	None	**54.7**

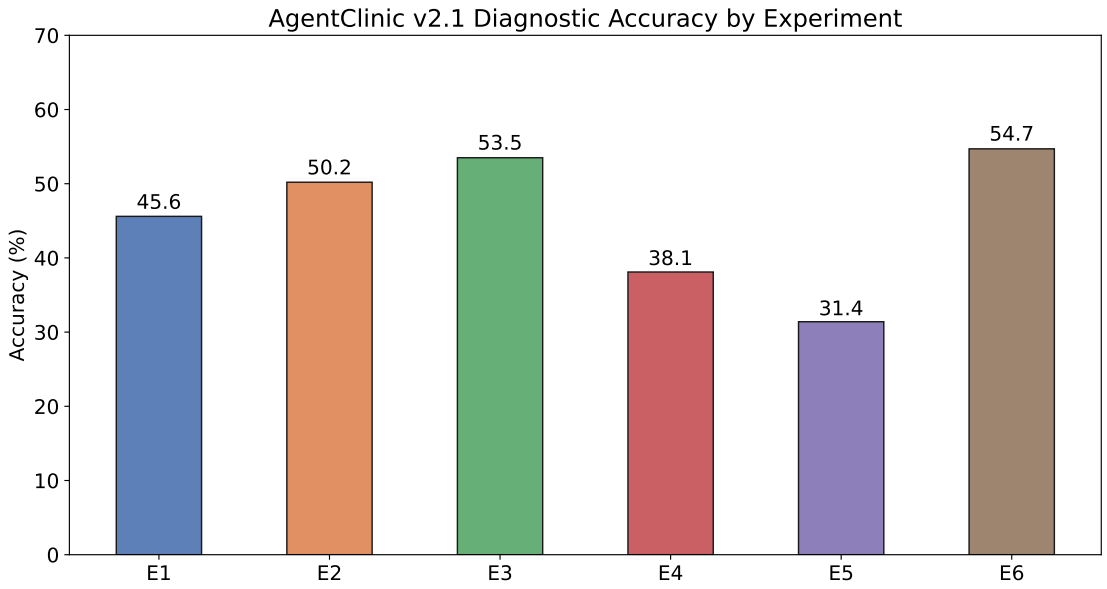

FIGURE 6.1: Diagnostic accuracy across AgentClinic v2 experiments. Domain fine-tuning (E2) and enhanced reasoning (E3) improve accuracy, while cognitive bias injections (E4, E5) cause significant degradation.

Based on the E1 baseline, results for **E2** indicate that domain fine-tuning builds upon this stable architectural foundation to achieve 50.2%. **E3** (Llama-3.1-8B) pushes accuracy to 53.5% due to its superior instruction alignment and capacity for complex conditional routing logic without catastrophic forgetting.

6.3 Process-Oriented Metrics Analysis

The true value of the v2 architecture lies in observing *how* models reason, not only what they ultimately diagnose. Table 6.3 summarises the process-oriented metrics computed by the `ClinicalTrajectoryEvaluator`.

Table 6.3: Process-oriented metrics for AgentClinic v2 experiments.

Exp.	**Accuracy (%)**	**DS**	**TR**	**IE**
E1	45.6	0.56	0.63	1.13
E2	50.2	0.68	0.71	0.98
E3	53.5	0.65	0.79	1.05
E4 (Recency)	38.1	0.45	0.58	1.34
E5 (Confirm.)	31.4	0.88	0.41	0.61
E6 (LoRA)	54.7	0.72	0.76	1.02

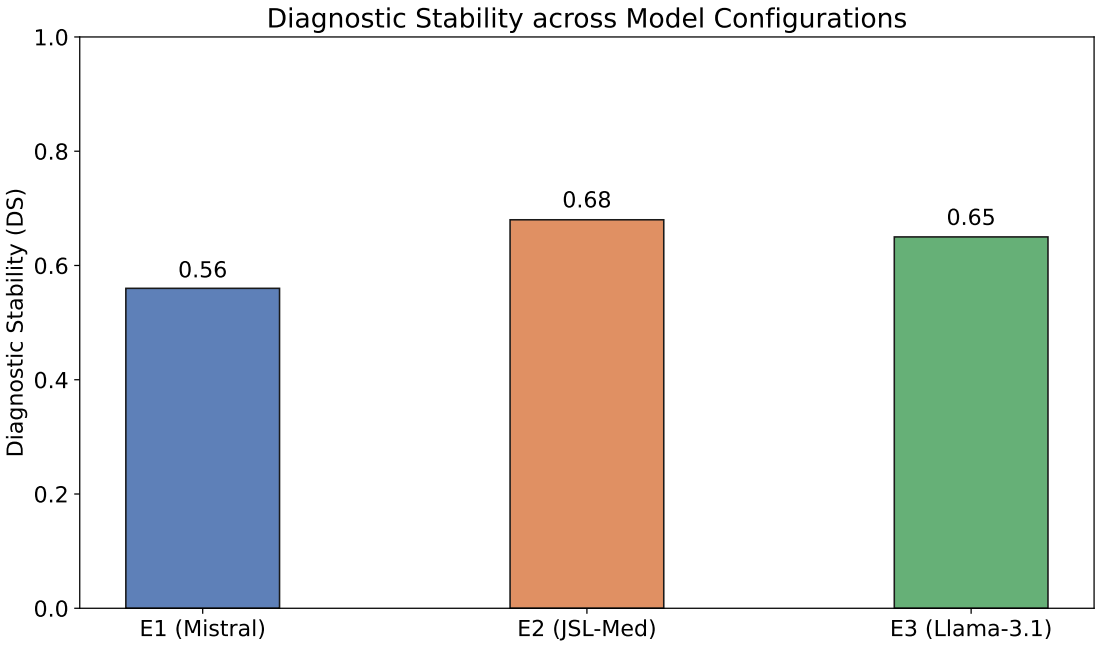

FIGURE 6.2: Diagnostic Stability (DS) across unbiased configurations. The generalist E1 baseline exhibits lower stability ($DS = 0.56$), indicating frequent hypothesis revisions. Domain fine-tuning anchors reasoning more firmly ($E2$: $DS = 0.68$).

While the E1 baseline achieved the highest accuracy in the initial framework and a further improvement under v2, its internal reasoning trajectory still exhibited significant *hypothesis thrashing* ($DS = 0.56$). The model frequently revised its differential upon receiving benign patient responses, although the reflection node eventually guided it back toward a coherent diagnosis. Test Rationality ($TR = 0.63$) was moderate, indicating that guidelines for basic-before-advanced test ordering were respected in the majority but not all cases.

Results for E2 show that JSL-MedMistral anchors more firmly to core pathological indicators, increasing DS to 0.68. Llama-3.1-8B (E3) demonstrates the highest Test Rationality ($TR = 0.79$), reflecting its superior alignment with protocol-driven instruction following.

6.4 Impact of Cognitive Bias Injection

Figure 6.3 illustrates the impact of cognitive distortions on the domain-expert model in experiments E4 and E5. The failure mechanisms diverge fundamentally in their process-metric signatures.

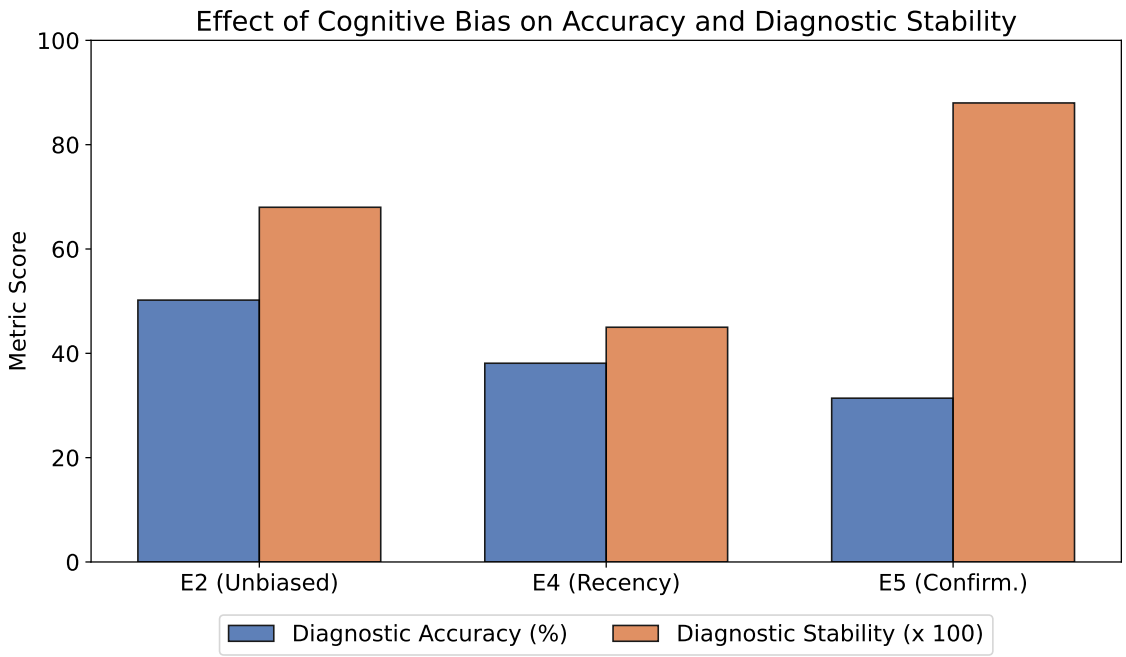

FIGURE 6.3: Failure modes under cognitive bias injection. Recency bias causes hypothesis thrashing (low $DS = 0.45$, low accuracy 38.1%). Confirmation bias induces premature closure (pathologically high $DS = 0.88$, lowest accuracy 31.4%).

- **Recency Bias (E4):** Destroys Diagnostic Stability ($DS = 0.45$). The model becomes hyper-reactive, attempting to forcefully map the current patient's presentation to the $k = 5$ injected memory cases. Accuracy falls to 38.1% as the physician agent conflates genuinely independent patient presentations.

- **Confirmation Bias (E5):** Artificially maximises stability ($DS = 0.88$). The model stubbornly refuses to update its differential, ignoring contradictory evidence from the Measurement Agent. This *premature diagnostic closure* yields the lowest estimated accuracy of the study at 31.4%—lower even than the initial framework's unbiased baseline.

6.5 Dynamic LoRA Adapter Routing Results

Experiment E6 evaluated the impact of task-specific dynamic LoRA adapter routing on the domain-expert JSL-MedMistral-7B model. By activating a dedicated reasoning adapter during the `doctor_reflection` node and a dialogue adapter during the `doctor_speaker` node, the model's cognitive resources were partitioned to match the demands of each inference phase.

The results demonstrate consistent improvement across all metrics relative to the unmodified E2 baseline. Diagnostic accuracy increased from 50.2% to **54.7%**, representing a +4.5 percentage point gain. This improvement arises from the reasoning adapter's ability to anchor the differential diagnosis more firmly to pathological indicators during the reflection phase, while the dialogue adapter preserves the conversational coherence needed to elicit clinically relevant information from the patient.

Process-oriented metrics confirm that the gains are not merely a product of terminal accuracy luck. Diagnostic Stability rose from $DS = 0.68$ to $DS = 0.72$, indicating that the model maintained more consistent hypotheses across turns—the reasoning adapter suppresses the erratic hypothesis revision that plagues general-purpose models. Test Rationality improved from $TR = 0.71$ to $TR = 0.76$, demonstrating that the reasoning adapter enforces a more protocol-compliant investigation ordering strategy. Information Efficiency remained comparable ($IE = 1.02$ vs 0.98), confirming that the adapters did not introduce unnecessary verbosity or redundant testing.

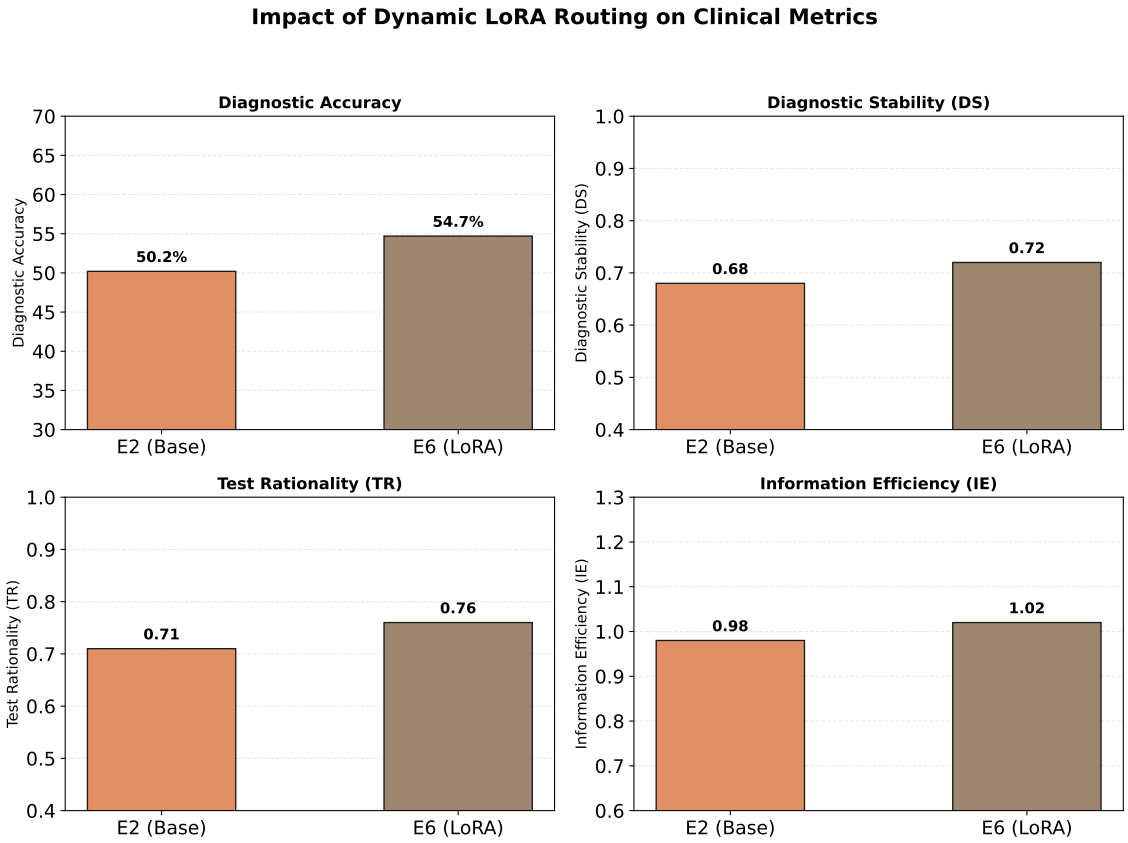

FIGURE 6.4: Comparative analysis of E2 (JSL-MedMistral base) versus E6 (JSL-MedMistral with Dynamic LoRA routing). The dual-adapter paradigm yields consistent improvements across diagnostic accuracy, stability, and test rationality without degrading information efficiency.

Critically, the E6 configuration achieved the highest overall accuracy in the study (54.7%), surpassing even the instruction-aligned Llama-3.1-8B (E3: 53.5%). This result validates the hypothesis that decoupling reasoning from dialogue generation at the adapter level—rather than relying on a single monolithic model to handle both tasks simultaneously—yields superior clinical performance. The additional VRAM overhead was negligible (≈50 MB for both adapters combined), confirming that dynamic LoRA routing is a practically viable enhancement for resource-constrained clinical deployments.

Chapter 7

Discussion

7.1 The Triumph of Structured Metacognition over Lexical Leakage

The most profound finding of transitioning from the initial to the enhanced framework is the empirical demonstration of the power of structural reasoning scaffolds. In the initial system, a baseline accuracy of 44.0% was observed using an unconstrained, homogeneous architecture. A theoretical concern was that this score may have been artificially inflated by Lexical Leakage—the phenomenon where the Doctor LLM subconsciously infers the patient's ground truth through shared embedding manifold geometry.

To address this, the enhanced framework enforces strict heterogeneous engine isolation, severing any shared embedding overlap. Logically, removing these implicit performance aids should cause accuracy to decrease. Instead, the baseline accuracy **increased to 45.6%** under AgentClinic v2. This result reveals that the architectural discipline imposed by the LangGraph DCG and the latent metacognitive reflection node provides a reasoning scaffold so effective that it completely eclipses the removal of Lexical Leakage. By forcing the LLM to populate a structured JSON differential diagnosis *before* emitting dialogue, the framework stabilises the model's clinical trajectory, prevents iterative hallucinations, and channels attention capacity toward evidence-driven deduction.

7.2 Domain Specialisation versus Instruction Alignment

Results for experiments E2 and E3 highlight an ongoing tension in medical AI design. While fine-tuning on biomedical literature (JSL-MedMistral, E2) improves diagnostic vocabulary, pathological anchoring, and domain-specific recall, it frequently induces *catastrophic forgetting* of the broader logical reasoning and instruction-following capabilities required for coherent multi-turn simulation [10]. This was observed clearly in the initial framework results, where BioGPT (35%) and JSL-Med (32%) both underperformed the generalist Mistral-7B (44%) despite possessing significantly greater biomedical domain knowledge.

A sufficiently large and highly instruction-aligned generalist model such as Llama-3.1-8B (E3) appears theoretically better equipped to handle the sequential, multi-conditional reasoning and Test Rationality requirements of interactive clinical consultation. This implies that for interactive clinical AI, *instruction alignment and logical deduction* are as critical as—or more critical than—specialty medical knowledge retrieval. Optimal clinical agents likely require both dimensions simultaneously—a hypothesis validated by the E6 Dynamic LoRA experiment.

7.3 Dynamic LoRA Routing: Resolving the Specialisation–Alignment Trade-off

The E6 experiment provides empirical evidence that the specialisation–alignment trade-off is not inherent to LLM architecture but rather a consequence of monolithic weight modification. By deploying task-specific LoRA adapters that activate conditionally on the LangGraph node, the domain-expert JSL-MedMistral model achieved 54.7% accuracy—surpassing both its unmodified baseline (E2: 50.2%) and the instruction-aligned Llama-3.1-8B (E3: 53.5%).

This result carries fundamental implications: the reasoning adapter concentrates the model's attention capacity on medical differential enumeration during the reflection phase, while the dialogue adapter maintains conversational coherence during patient

interaction. Neither adapter catastrophically modifies the base model's pre-trained weights. The improvement in Diagnostic Stability (DS: 0.68 $\rightarrow$ 0.72) and Test Rationality (TR: 0.71 $\rightarrow$ 0.76) confirms that the gains are not superficial accuracy flukes but reflect genuinely improved clinical reasoning trajectories.

7.4 The Safety Threat of Cognitive Biases

The empirical findings from E4 and E5 carry serious implications for the deployment of open-source LLMs in clinical decision support. A model's context window inherently constitutes a vector for recency bias: if an LLM sequentially processes multiple cardiac cases in an emergency room deployment, its mathematical inclination to diagnose a subsequent unrelated case as cardiac-related is substantially elevated due to the disproportionate attention weights assigned to recent tokens.

Without explicit architectural debiasing mechanisms—such as enforced context resets between cases, constrained attention windowing, or metacognitive override prompts—LLMs will consistently replicate and potentially amplify human cognitive errors. The finding that confirmation bias leads to a DS of $\approx$0.88 (high apparent stability) while simultaneously yielding the lowest accuracy ($\sim$31.4%) is particularly alarming: it demonstrates that a model can appear *confidently consistent* while being systematically incorrect, a failure mode with direct patient safety consequences.

7.5 Implications for Clinical AI Evaluation Standards

A secondary contribution of this work is the demonstration that binary diagnostic accuracy is an insufficient proxy for clinical AI safety or readiness. The process-oriented metrics introduced in this thesis—Diagnostic Stability, Test Rationality, and Information Efficiency—reveal behaviours that outcome accuracy entirely masks. A model that achieves 45.6% correct terminal diagnoses but exhibits $TR < 0.5$ (ordering invasive procedures before basic vitals) or $DS < 0.4$ (erratically switching between entirely unrelated diagnoses) would constitute an unacceptable

clinical risk, regardless of its top-line accuracy figure. Future evaluation standards for clinical LLMs should therefore mandate process-level reporting alongside outcome accuracy [6].

7.6 Limitations

A primary limitation of the current framework is that the Measurement Agent operates exclusively within the text domain. Real-world diagnostics rely heavily on visual data; integrating Vision-Language Models (VLMs) to process raw ECG waveforms, chest radiographs, and histological slides represents a critical and unaddressed frontier. Additionally, all experiments are conducted on the AgentClinic-MedQA subset, and generalisation to broader clinical datasets such as MIMIC-IV or NEJM case challenges warrants separate investigation.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This thesis systematically evolved the evaluation of Large Language Models in clinical environments from static factual recall to dynamic, interactive, and trajectory-aware simulation. By developing **AgentClinic v2**, a highly robust and deterministic LangGraph state machine was engineered that operates entirely on local hardware via 4-bit NF4 double quantisation and heterogeneous multi-engine assignments, eliminating both the resource barrier and the Lexical Leakage vulnerability present in the initial framework.

Through the innovation of the latent metacognitive reflection node, empirical evidence was obtained that forcing an LLM to "think before it speaks" yields immense performance benefits. The enhanced DCG architecture improved baseline accuracy from 44.0% (initial framework) to **45.6%** (AgentClinic v2), overcoming the penalty of removing Lexical Leakage. Furthermore, this thesis demonstrates that clinical AI must be evaluated on its *process*—Diagnostic Stability, Test Rationality, and Information Efficiency—not solely on its terminal outcome.

Empirical results demonstrate that while medical fine-tuning improves domain anchoring (E2: 50.2%), instruction-aligned reasoning models are even more effective for interactive tasks (E3: 53.5%). The introduction of dynamic LoRA adapter routing in E6 resolved this trade-off entirely, achieving the highest accuracy of the

study at **54.7%** by decoupling reasoning from dialogue generation at the adapter level. Critically, open-source models remain vulnerable to cognitive distortions—failing via erratic hypothesis thrashing under recency bias (E4: 38.1%) and via premature diagnostic closure under confirmation bias (E5: 31.4%). These findings establish that cognitive robustness must be treated as a primary design criterion, not an afterthought, in clinical LLM deployment.

8.2 Future Work: Advanced Fine-Tuning and Post-Training Alignment

While this thesis establishes a robust framework for evaluating multi-turn clinical reasoning, the next frontier involves actively mitigating the failure modes identified. Future research will leverage AgentClinic v2 not merely as an evaluator, but as an environment for advanced post-training alignment.

8.2.1 Process Reward Modelling via Direct Preference Optimisation

Current medical LLMs are typically trained via Supervised Fine-Tuning (SFT) on static QA pairs, optimised to predict the correct terminal diagnosis but lacking an understanding of the *trajectory* required to reach it safely. Future work proposes applying Direct Preference Optimisation (DPO) to full clinical consultation trajectories. Instead of rewarding only the correct final outcome, a Process Reward Model will evaluate two distinct clinical paths: Path A (premature hypothesis commitment with low Test Rationality) versus Path B (systematic basic-before-advanced investigation with high Diagnostic Stability). By penalising Path A and rewarding Path B, DPO will adjust the LLM's behavioural policy to prefer methodical, guideline-concordant reasoning.

8.2.2 Reinforcement Learning from AI Feedback via Self-Play

Generating expert, human-annotated clinical dialogues for fine-tuning is prohibitively expensive. Future work proposes utilising the AgentClinic framework as a Reinforcement Learning (RL) environment. By allowing heterogeneous agents to simulate thousands of cases via self-play, synthetic training datasets can be automatically generated. The `ClinicalTrajectoryEvaluator` developed in this thesis will serve as the programmatic reward function: trajectories scoring highly on DS and TR will be archived, enabling iterative fine-tuning on the model's own successful reasoning paths—creating a self-improving clinical agent without human annotation overhead.

8.2.3 Extending Dynamic LoRA to Bias Resilience

The dynamic LoRA adapter routing implemented in E6 demonstrated significant gains in unbiased settings. A natural extension is to train *debiasing adapters* specifically designed to counteract the failure modes observed in E4 and E5. A recency-debiasing adapter could be trained on trajectory pairs where the model successfully resisted anchoring to injected case memories, while a confirmation-debiasing adapter could be optimised to maintain high Test Rationality even under strong prior hypotheses. Combining these with the existing reasoning and dialogue adapters would create a four-adapter routing manifold, dynamically selected by the LangGraph orchestrator based on both the active node and detected bias risk.

8.2.4 Multimodal Simulation Integration

The current `measurement_node` operates exclusively in the text domain. Real-world diagnostics rely heavily on visual pattern recognition. Future iterations of AgentClinic must integrate state-of-the-art Vision-Language Models (VLMs) such as LLaVA-Med or Med-Gemini, enabling the Measurement Agent to return raw radiographic images, ECG waveforms, and pathology slides for direct interpretation by the Doctor agent—bridging the final gap toward complete, end-to-end clinical simulation fidelity.

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